

Introduction to cheminformatics

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Definition

Combines chemistry, computer science, and information science to organize, process, and present chemical data.



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Combines chemistry, computer science, and information science to organize, process, and present chemical data.

Computers don't know chemistry, the translation of “chemistry” to computer language is necessary.



It means **S**implified **M**olecular **I**nput **L**ine **E**ntry **S**ystem.

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A way to write molecules in one string that is easy for humans to read.

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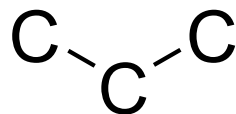
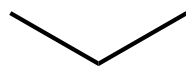
A way to write molecules in one string that is easy for humans to read.

Examples:

It means **S**implified **M**olecular **I**nterface **L**anguage **E**ntering **S**ystem.

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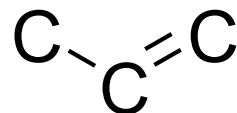
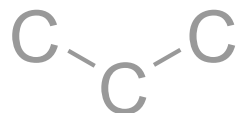
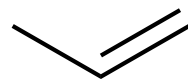
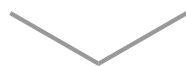
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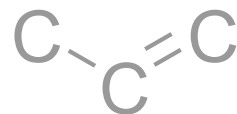
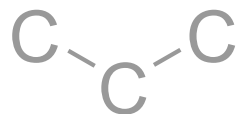
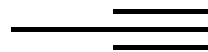
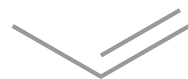
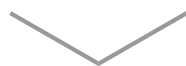
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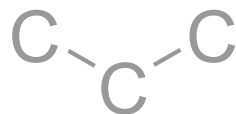
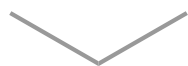
Examples:



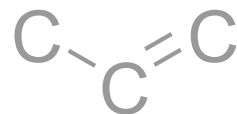
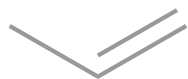
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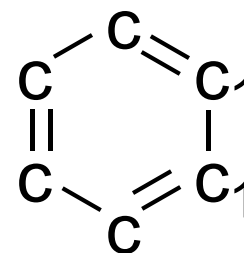
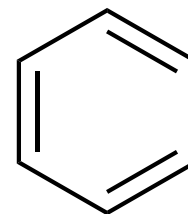
CCC



CC=C



CC#C

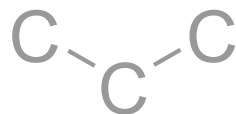
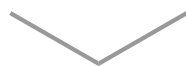


c1ccccc1

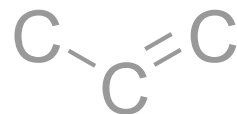
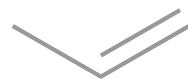
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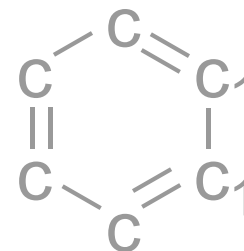
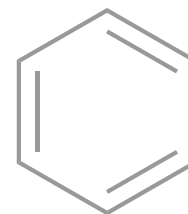
CCC



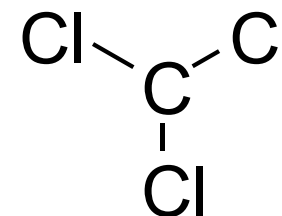
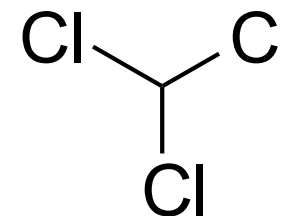
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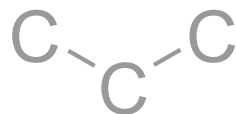
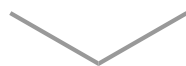


ClC(Cl)Cl

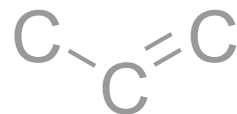
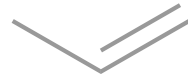
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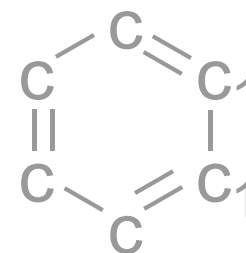
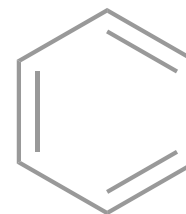
CCC



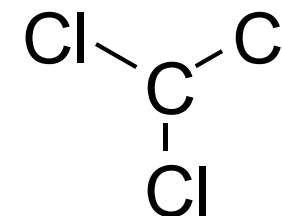
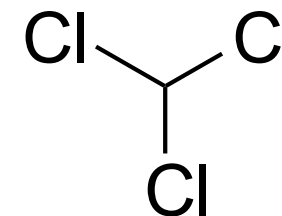
CC=C



CC#C



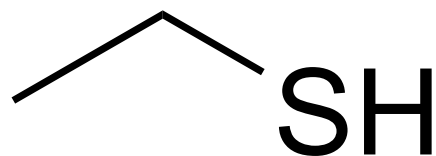
c1ccccc1



ClC(Cl)Cl

Canonical SMILES

SMILES of molecule?

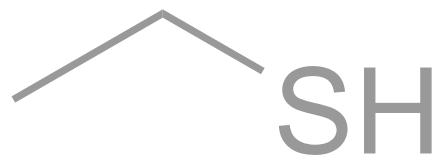


a) CCS

b) C(S)C

c) SCC

SMILES of molecule?



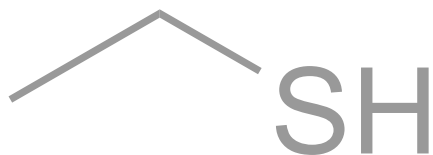
a) CCS

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All are correct, but in a database, will lead to duplicates.

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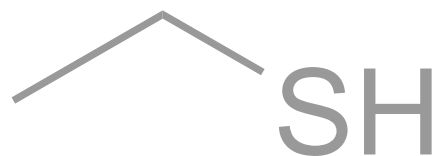
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The canonical SMILES must be obtained to get a *unique standardized text representation* is
a) CCS.

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You can check with this [tool](#).